NES AI FINANCE & INSURANCE INTELLIGENCE PLATFORM

Backtesting Methodology Specification

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Build approach: Cline-assisted development • Local-first Windows application

Scope: Personal research, paper trading and insurance document assistance

Assumptions, metrics, bias controls and validation

1. Purpose

Define the minimum methodology for historical strategy testing so results are reproducible and do not accidentally overstate performance.

2. Required inputs

- Strategy name/version and parameters.
- Symbol or universe and exchange.
- Start/end dates and bar interval.
- Data source, timestamp convention and adjustment policy.
- Starting capital and position-sizing method.
- Fees, taxes and slippage assumptions where relevant.
- Order timing and fill assumptions.
- Benchmark and risk-free assumptions if used.

3. Methodology rules

- Avoid look-ahead bias: a decision may only use information available at that time.
- Avoid survivorship bias where universe history is available and relevant.
- Document missing bars, stale values and corporate-action treatment.
- Make signal timing and fill assumptions explicit.
- Use out-of-sample or walk-forward evaluation when appropriate.
- Do not tune parameters on the final evaluation period without disclosure.
- Separate training, validation and test periods for ML workflows.

4. Required outputs

Metric	Definition requirement
Total return	Specify cash flows, fees and calculation period.
Benchmark return	Name benchmark and align periods.
Maximum drawdown	Document equity curve basis and peak-to-trough method.
Trade count	Define what counts as an opened/closed trade.
Win rate	Define winning trade and handling of open positions.
Costs	Show fee and slippage assumptions.
Equity curve	State whether returns are realised, mark-to-market or both.

5. Validation tests

- Use tiny hand-calculated datasets to verify trade timing and metrics.
- Test missing bars, gaps, duplicate bars and extreme values.
- Test no-trade and single-trade scenarios.
- Check that the same data and parameters reproduce the same result.
- Compare a simple baseline strategy against expected output.

6. Presentation rules

- Label results historical, not predictive.
- Show test period, source, assumptions and warnings near the results.
- Do not use animated count-ups to obscure negative performance.
- Do not present model confidence as calibrated unless calibration has been evaluated.

Backtest results are simulations. They are not proof of future performance and are not a guarantee of returns.